

Semi-Empirical Noise Source Prediction Models

for Turbofan-Powered Aircraft

Fan (Heidmann) · Jet Mixing (Lighthill/Stone) · Core/Combustor (GE–SAE) · Airframe (Fink)

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Abstract

This document summarizes the four classical semi-empirical noise-source prediction models that underlie NASA’s Aircraft Noise Prediction Program (ANOPP) and equivalent industry tools. These models predict the *source* spectrum and directivity of aircraft noise components before any atmospheric propagation, geometric spreading, or A-weighting is applied – i.e. they answer the question “how loud is the source itself, and in which direction, at which frequency?” rather than “how loud does it sound on the ground at a given distance?”. The four sources covered are: fan/compressor noise (Heidmann’s interim method), jet mixing noise (Lighthill’s similarity law and Stone’s refined semi-empirical model), core/combustor noise (the GE/SAE direct-combustion-noise model), and airframe noise (Fink’s component method for clean wing, slats, flaps, and landing gear).

1 Why These Models Exist

An aircraft’s total noise signature is the incoherent (energy) sum of several physically distinct noise-generating mechanisms, each dominant in a different flight regime:

- **Jet mixing noise** dominates at high thrust (takeoff), because it scales with a very high power of jet velocity.
- **Fan noise** is significant across the whole flight envelope and often dominates at approach power, radiating both broadband and discrete-tone (buzz-saw, blade-passage) components.
- **Core/combustor noise** is a low-frequency, comparatively minor but non-negligible contributor, especially at low power (approach/idle) where jet noise has dropped away.
- **Airframe noise** (clean wing, high-lift devices, landing gear) is purely aerodynamic – it exists even with the engines at idle – and becomes the dominant source during approach, when thrust is low but flaps, slats and gear are extended.

Each model below predicts, for one component, the mean-square acoustic pressure $\overline{p^2}$ (or an equivalent sound pressure level in dB) as a function of frequency f and observer angle (θ, ψ) , referenced to a distance r or H . Component predictions are then summed energetically ($10 \log_{10} \sum 10^{L_i/10}$) to obtain the total source spectrum, which is subsequently propagated to the ground (spherical spreading, atmospheric absorption, Doppler/convective amplification) exactly as in the propagation formulas discussed earlier.

2 Fan and Compressor Noise — Heidmann’s Interim Method

2.1 Background

Developed by M. F. Heidmann (NASA TM X-71763, 1979), this remains the standard fan/compressor noise module in ANOPP. It treats the fan as producing five distinct noise components: inlet broadband, inlet discrete tones (blade-passage harmonics), inlet combination tones ("buzz-saw" noise from supersonic relative tip Mach numbers), discharge broadband, and discharge discrete tones.

2.2 Governing equations

Characteristic (peak) sound pressure level, common to all components:

$$L_c = 20 \log_{10} \left(\frac{\dot{m}}{\dot{m}_0} \right) + 10 \log_{10} \left(\frac{\Delta T}{\Delta T_0} \right) + F_1(M_{TR}, M_{TR,D}) + F_2(RSS) + F_3(\theta) [+C_{IGV}]$$

Spectral distribution (broadband, log-normal shape about the blade-passage frequency):

$$SPL(f) = L_c + F_4 \left(\frac{f}{f_b} \right), \quad F_4 = 10 \log_{10} \exp \left[-\frac{1}{2} \left(\frac{\ln(f/2f_b)}{\ln \sigma} \right)^2 \right], \quad \sigma = 2.2$$

Discrete-tone (blade-passage) harmonic levels relative to L_c :

$$\text{No inlet guide vanes: } L_k = 3 - 3k \quad \text{With inlet guide vanes: } L_1 = 0, \quad L_k = -3 - 3k \quad (k \geq 2)$$

Tip-Mach "cut-off" correction (fundamental tone is attenuated when it cannot propagate):

$$\delta = \frac{M_T \cdot N_v}{B} \quad \text{if } \delta \leq 1.05 \Rightarrow \text{reduce fundamental (k=1) tone by 8 dB}$$

Combination-tone (buzz-saw) noise, first-stage fans only, when $M_{TR} > 1.0$: the same L_c expression is evaluated (with its own F_1, F_3 and $C_{IGV} = -5$ dB) at three sub-harmonic spectral centers $f/f_b = \frac{1}{2}, \frac{1}{4}, \frac{1}{8}$, and the three resulting spectra are summed energetically.

Rotor-stator spacing correction:

$$\Delta L_{RSS} = -5 \log_{10} \left(\frac{RSS}{300} \right) \quad (\text{broadband}) \quad \Delta L_{RSS} = -10 \log_{10} \left(\frac{RSS}{300} \right) \quad (\text{discrete tones})$$

2.3 Parameter definitions

Symbol	Meaning
$\dot{m}$	Mass flow rate through the fan/compressor face [kg/s]
$\dot{m}_0$	Reference mass flow, 0.453 kg/s (1 lbm/s)
ΔT	Total temperature rise across the stage [K]
ΔT_0	Reference temperature rise, 0.555 K (1°R)
M_{TR}	Rotor tip relative inlet Mach number (operating point)
$M_{TR,D}$	Rotor tip relative inlet Mach number at the design point
M_T	Rotor tip (wheel-speed) Mach number
N_v	Number of stator vanes
B	Number of rotor blades
RSS	Rotor-stator axial spacing, as a percentage of rotor chord
θ	Polar directivity angle measured from the fan/compressor axis [deg]
f_b	Blade-passage frequency, $f_b = B \cdot \text{RPM}/60$ [Hz]
k	Harmonic order of the blade-passage tone ($k = 1, 2, 3, \dots$)
C_{IGV}	Correction for inlet guide vanes or 2nd-stage fans: +3 dB (discharge broadband), +6 dB (discharge tone), -5 dB (combination tone), 0 otherwise
F_1, F_2, F_3, F_4	Empirical functions (Mach number, spacing, directivity, spectrum shape) read from Heidmann's reference curves/tables

Source: M. F. Heidmann, "Interim Prediction Method for Fan and Compressor Source Noise," NASA TM X-71763 (1979).

3 Jet Mixing Noise — Lighthill's Law and Stone's Model

3.1 Background

Jet mixing noise is generated by turbulent shear-layer mixing downstream of the nozzle. Its theoretical foundation is Lighthill's acoustic analogy (1952), which shows that for an unheated, low-speed turbulent jet the total radiated acoustic power scales with the *eighth power* of the jet velocity – by far the steepest velocity dependence of any aircraft noise source, which is why jet noise dominates only at high thrust and falls away rapidly as power is reduced.

3.2 Lighthill's similarity law

$$W \propto K \rho_\infty \frac{D_j^2 V_j^8}{c_\infty^5} \implies OASPL \propto 80 \log_{10} \left(\frac{V_j}{c_\infty} \right) + 20 \log_{10} \left(\frac{D_j}{r} \right) + 20 \log_{10} \left(\frac{\rho_j}{\rho_\infty} \right) + \text{const.}$$

3.3 Stone's semi-empirical refinement (used in ANOPP / SAE ARP876)

Stone's model (as implemented in ANOPP's jet-noise module, ST2JET) splits a modern turbofan's exhaust into up to three coherent mixing regions – the outer-stream shear layer, the inner-stream shear layer, and the fully merged jet – each treated as its own virtual noise source with its own velocity exponent, spectrum, and directivity, then combined energetically:

$$\overline{p_m^2}(f, \theta) = \overline{p_{ref,m}^2} \cdot D_m(\theta) \cdot F_m(S_m, \theta) \cdot H_m(M_\infty, \theta) \cdot \frac{G_c G_p}{r^2}, \quad m = \text{outer, inner, merged}$$

with the reference mean-square pressure of each stream component of the classical form

$$\overline{p_{ref,m}^2} = 2.502 \times 10^{-6} \left(\frac{\rho_j}{\rho_\infty} \right) \left(\frac{V_j}{c_\infty} \right)^{7.5-8} \frac{A_j}{r_{ref}^2} \rho_\infty^2 c_\infty^4$$

3.4 Parameter definitions

Symbol	Meaning
V_j	Fully-expanded jet exhaust velocity [m/s]
D_j, A_j	Effective nozzle exit diameter [m] and area [m ²]
ρ_j, ρ_∞	Jet exhaust density and ambient air density [kg/m ³]
c_∞	Ambient speed of sound [m/s]
M_∞	Aircraft (flight) Mach number
r, r_{ref}	Observer distance and reference distance for normalization [m]
$S_m = f D_j / V_j$	Strouhal number of mixing region m
$D_m(\theta)$	Directivity function of mixing region m (peaks near 30° from jet axis)
$F_m(S_m, \theta)$	Normalized spectrum shape function
H_m	Forward-flight (relative velocity) correction factor
G_c, G_p	Configuration/geometry correction factors (e.g. nozzle coplanar/mixed exhaust)
K	Dimensional/empirical proportionality constant

Sources: M. J. Lighthill, "On Sound Generated Aerodynamically," Proc. Roy. Soc. A 211 (1952); J. R. Stone jet-noise module as documented in SAE ARP876 ("Gas Turbine Jet Exhaust Noise

Prediction”) and the ANOPP Theoretical Manual. The exact numerical coefficients, velocity exponent per stream, and tabulated D_m/F_m functions are given in SAE ARP876F; the value above is the widely-cited reference form.

4 Core (Combustor) Noise — the GE/SAE Model

4.1 Background

Core noise (direct combustion noise) is generated by unsteady heat release and pressure fluctuations in the combustor, partially attenuated as it passes through the turbine stages before radiating from the exhaust. It is a low-frequency, low-level source compared to jet or fan noise, but becomes noticeable at low power settings (approach/idle) where the louder sources have receded.

4.2 Governing equation

Total radiated acoustic power:

$$\Pi = 10^{K/10} c_0^2 \dot{m}_c \left(\frac{T_{t,ce} - T_{t,ci}}{T_{t,ci}} \right)^2 \left(\frac{P_{t,ci}}{P_0} \right)^2 F_{TA}$$

Turbine attenuation factor:

$$F_{TA} = \left(\frac{\Delta T_{des}}{T_0} \right)^{-4}$$

Far-field mean-square pressure (directivity and spectrum applied separately, per SAE ARP876 normalization):

$$\overline{p^2}(f, \theta) = \frac{\Pi \rho_0 c_0^2}{4\pi r^2} D(\theta) S\left(\frac{f}{f_p}\right)$$

4.3 Parameter definitions

Symbol	Meaning
Π	Total radiated acoustic power of the core/combustor [W]
K	Empirical constant: -60.53 dB (standard-size engines), -64.53 dB (small engines)
$\dot{m}_c$	Combustor mass flow rate [kg/s]
$T_{t,ci}, T_{t,ce}$	Total temperature at combustor inlet and exit [K]
$P_{t,ci}$	Total pressure at combustor inlet [Pa]
c_0	Reference (sea-level) speed of sound, 340.294 m/s
P_0, T_0	Reference sea-level pressure (101.325 kPa) and temperature (288.15 K)
ΔT_{des}	Design-point temperature drop across the turbine stage(s) [K]
F_{TA}	Turbine attenuation factor (accounts for acoustic energy lost crossing the turbine)
r	Observer distance [m]
θ	Polar directivity angle [deg]
f_p	Peak (characteristic) frequency of the core noise spectrum [Hz]
$D(\theta), S(\cdot)$	Normalized directivity and spectrum shape functions (SAE ARP876 tables)

Source: GE/SAE direct combustion-noise model, as implemented in ANOPP and documented in SAE ARP876 and NASA’s core-noise literature (e.g. Mathews & Rekos correlation lineage).

5 Airframe Noise — Fink’s Component Method

5.1 Background

Unlike the three engine-noise sources above, airframe noise is purely aerodynamic: it is generated by turbulent flow over the clean airframe, the high-lift devices (leading-edge slats, trailing-edge flaps), and the landing gear. M. R. Fink’s method (FAA-RD-77-29, 1977, adopted into ANOPP) treats each of these as an equivalent compact noise source (mostly dipole-like), each with its own reference acoustic power law, geometric scaling, spectrum, and directivity, amplified by the same convective Doppler factor that appears whenever a source moves relative to the observer.

5.2 General normalized form (common to all components)

$$\overline{p^2}(f, \theta, \psi) = \frac{\Pi^* \rho_\infty c_\infty^2}{4\pi H^2} D(\theta, \psi) F(S) \frac{1}{(1 + M_\infty \cos \theta)^4}, \quad \Pi^* = K M_\infty^a G(\text{geometry})$$

The $(1 + M_\infty \cos \theta)^{-4}$ factor is the classical convective amplification of a compact dipole source moving at flight Mach number M_∞ – it makes the noise louder ahead of the aircraft’s direction of travel (as heard on the ground) than a stationary source would be.

5.3 Clean wing (trailing-edge) noise

$$\overline{p^2}(f, \theta, \psi) = \frac{K \rho_\infty^2 V_\infty^5 \delta_w b}{4\pi H^2 a_\infty} F[St(f, \theta)] \frac{D(\theta, \psi)}{(1 + M_\infty \cos \theta)^4}, \quad K = 7.075 \times 10^{-6}$$

The V_∞^5 scaling (rather than the jet’s V^8) is characteristic of a dipole source (turbulent boundary layer scattering at the trailing edge), which radiates far less efficiently per unit velocity increase than the jet’s quadrupole mixing noise.

5.4 Leading-edge slat noise

Modeled as the incoherent sum of two sources: (i) a clean-wing-like turbulent trailing-edge term evaluated with the slat’s own chord and span, and (ii) an additional broadband component (roughly 3–6 dB above the first) attributed to unsteady flow reattachment in the slat cove. Same V_∞^5 and Doppler-factor structure as the clean wing, but referenced to slat geometry instead of the full wing.

5.5 Trailing-edge flap noise

$$\text{Same normalized form, with source strength} \propto \rho_\infty^2 V_\infty^{5-6} (\text{flap span}) \times f(\delta_f)$$

Flap noise is dominated by (a) trailing-edge scattering along the flap span, scaling similarly to the clean wing, and (b) side-edge noise at the flap tips (vortex shedding around the flap edge), which grows strongly with flap deflection angle δ_f and can dominate at large flap settings – this is why deploying flaps further generally increases approach noise even though it allows a lower approach speed.

5.6 Landing gear noise

$$\text{Same normalized form, source strength} \propto \rho_\infty^2 V_\infty^6 n_w d^2 (+ \text{strut contribution})$$

Modeled as a set of bluff-body (strut and wheel) dipole sources; each gear’s contribution depends on the number of wheels n_w , wheel diameter d , strut length and count, with main and nose gear treated separately using different geometry factors and directivities. Because of the V_∞^6 scaling and the fact that gear is normally extended only at low altitude/low speed near the runway, it is a dominant approach-noise contributor together with flaps.

5.7 Parameter definitions

Symbol	Meaning
V_∞	Freestream (aircraft) velocity [m/s]
M_∞	Freestream (flight) Mach number
ρ_∞, a_∞	Ambient air density [kg/m ³] and speed of sound [m/s]
δ_w	Boundary-layer thickness at the trailing edge [m]
b	Wing span (or relevant component span) [m]
H	Distance from source to observer [m]
θ, ψ	Polar and azimuthal directivity angles [deg]
$St(f, \theta)$	Strouhal number based on local chord/thickness and V_∞
$F(S)$	Normalized spectrum shape function
$D(\theta, \psi)$	Normalized directivity function
δ_f	Flap deflection angle [deg]
n_w	Number of wheels on the gear assembly
d	Wheel diameter [m]
K, a	Empirical constant and Mach exponent, component-specific
$G(\cdot)$	Geometry function specific to each component (area, span, wheel/strut count, etc.)

Source: M. R. Fink, "Airframe Noise Prediction Method," FAA-RD-77-29 (1977), as implemented in the ANOPP Theoretical Manual (NASA TM-83199, Part 2). **Note on precision:** the clean-wing formula and its constant $K = 7.075 \times 10^{-6}$ are reproduced directly from the primary literature. For the slat, flap, and landing-gear sub-models, the functional form and physical scaling shown above (velocity exponent, governing geometric parameters, Doppler factor) are well established in the literature, but the exact tabulated K , a , $D(\theta, \psi)$ and $F(S)$ values for each of those three components should be taken from Fink's original report or the ANOPP Theoretical Manual directly before use in a certified engineering calculation.

6 Summary — Velocity Scaling by Source

The steepness of each source's velocity dependence explains why the dominant noise source changes with flight condition (a theme directly relevant to the approach-noise / SEL calculation discussed earlier in this thread, where jet noise is negligible and fan + airframe dominate):

Source	Approx. velocity scaling	Dominant flight regime
Jet mixing noise	$V_j^{7.5-8}$	Takeoff / high thrust
Fan noise	Mach-number function $F_1(M_{TR})$, steep near $M_{TR} \approx 1$	All regimes; tones dominant
Core/combustor noise	$\propto (\Delta T/T)^2 (P_{t,ci})^2$, weak velocity dependence	Low power (approach/idle)
Airframe – clean wing/slat	V_∞^5	Approach (flaps/gear out, etc.)
Airframe – flap/gear	V_∞^6	Approach

7 References

1. Heidmann, M. F., *Interim Prediction Method for Fan and Compressor Source Noise*, NASA TM X-71763, 1979.
2. Lighthill, M. J., *On Sound Generated Aerodynamically. I. General Theory*, Proceedings of the Royal Society A, Vol. 211, 1952.
3. Stone, J. R., *Interim Prediction Method for Jet Noise*, as implemented in ANOPP (ST2JET module); SAE ARP876, *Gas Turbine Jet Exhaust Noise Prediction*.

4. Emmerling, J. J., Kazin, S. B., Matta, R. K., *Core Engine Noise Control Program*, and the GE/SAE combustion-noise model as documented in SAE ARP876 and NASA core-noise literature.
5. Fink, M. R., *Airframe Noise Prediction Method*, FAA-RD-77-29, Federal Aviation Administration, 1977.
6. Zorumski, W. E., *Aircraft Noise Prediction Program Theoretical Manual*, NASA TM-83199, Parts 1 & 2, 1982.